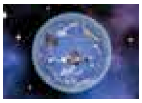
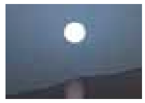


Simulate climate
This tool lets you simulate the climate on exoplanets, and tweak atmospheric variations in real time. **https://dgit.in/aug20-51**



Alien sunsets
NASA has made a series of simulations that shows what sunsets on other planets would look like. **https://dgit.in/aug20-52**

ATMOSPHERIC SCIENTIST

The ones who study the Earth's atmosphere and all its inner workings

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Weather affects everyone, everywhere. Therefore, there is a tangible need for professionals who extensively research and study the atmosphere of the Earth. The interdisciplinary field of atmospheric science allows for accurate predictions of weather and climate change, and identification of environmental threats. An atmospheric scientist, however, is much more than just a meteorologist or a weather forecaster. An atmospheric scientist is basically anyone who works in studying the atmosphere and the inner workings of our entire planet. The job is extremely research-intensive as well. So, let's get into everything you must know if you want to build a career as an atmospheric scientist.

WHAT DOES AN ATMOSPHERIC SCIENTIST DO?

To understand what an atmospheric scientist does, you must first know what the field of atmospheric science is. Atmospheric science is an interdisciplinary field of study, combining components of physics and chemistry, that ultimately focus on the structure and dynamics of the atmosphere of the Earth. Atmospheric scientists are usually knowledgeable in this particular field and go on to study the scientific and mathematical aspects of Earth's atmosphere,

Source: World View

A high-altitude weather balloon that pulled off a 5-day mission

climate and weather during the course of their careers.

The field of atmospheric science includes various areas of interest and an atmospheric scientist usually focuses on one or a few of these areas. As per NASA, the different areas of interest in atmospheric science include – climatology, dynamic meteorology, cloud physics, atmospheric chemistry, atmospheric physics, aeronomy (the study of the upper atmosphere) and oceanography. So, there are different types of atmospheric scientists based on their area of interest. There's another field in atmospheric science that deals with studying the atmospheres of the planets and the moons in our solar system.

Atmospheric scientists are tasked with studying the various areas of interest (as mentioned above) and figuring out how these dynamic conditions affect human activity and the Earth, in general. Atmospheric scientists may develop reports, forecasts, collect, analyse and compile data from the field; assist in the development of new data collection instruments; and even advise clients and governments about risks caused by climate conditions and climate change. They also thoroughly study older data to build a bigger and clearer picture of the climate, weather and, atmospheric conditions in the past and correlate the data points found today to the past to understand progress or decline better.

TOOLS OF THE TRADE

The typical tasks an atmospheric scientist is expected to be able to perform are the following – measuring temperature, air pressure and other properties of Earth's atmosphere, developing and using computer models, analysing meteorological data, producing weather maps and graphics, reporting weather conditions, using other computer and mathematical models, satellites and radar data, and more. In order to accomplish these tasks, a wide range of instruments and computer programs are used by atmospheric scientists. Some examples include weather bal-